

Functional genetic report · longevity

# **Andropause and Reproductive Ageing**

Sérgio F · Report date: 02-10-2026

As the genes condition the decrease of testosterone with age (andropause), SHBG, LH/FSH axis, loss of Y chromosome in blood cells and reproductive calendar. Only genes (chip + local imputation + polygenic scores).  
Complements the report "Hormones" without repeating the same SNPs.

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## Summary

**17**

SNPs analyzed

**15**

in the chip

**1**

imputed ( $DR2 \geq 0.8$ )

**1**

SNP not found in the data

**6**

Polygenic scores

In man, total testosterone decreases ~1% per year from the 35–40 years and free ~2%, because SHBG rises with age. Andropause (late-onset hypogonadism) depends mainly on adiposity, sleep, disease and drugs, but the genetic starting point counts: who starts out later reaches symptom thresholds. Another marker of male reproductive aging is the loss of Y chromosome in blood cells, linked to mortality, cancer and cardiovascular disease.

In summary: high testosterone genetic reserve: the testosterone score of man with 980 thousand variants is at the 89th percentile, with the genetic SHBG also high (the 87th percentile); therefore total testosterone tends to be high and free is closer to average. The loss of the Y chromosome is at the 37th percentile. Genetic puberty is late (the 88th percentile of the age of voice change).

### Favourable

- **Man's testosterone at the 89th percentile** (PGS003559, 980 thousand variants): more margins before andropause thresholds.
- **Loss of the mosaic Y chromosome at the 37th percentile** (114 variants): typical, low side.
- **SHBG rs12150660 GT**: 1 lower SHBG allele, which helps free testosterone.

### Unfavourable

- **High genetic SHBG** (the 87th percentile): with age SHBG rises further and "learns" testosterone (high SHBG protects glucose; see "Hormones").
- **TCL1A rs2887399 GG**: without the allele that protects against Y loss and clonal hematopoiesis; TERT rs7705526 heterozygous.
- **Not visible**: CAG repeat of the androgen receptor (receptor sensitivity) and CHEK2 1100delC.

**Interpretation:** genetics gives a testosterone reserve above average. Andropause, when it happens, will be mainly explained by modifiable factors: sleep, visceral adiposity, strength training and drugs (rapamycin can lower testosterone).

## How to read this report

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**Genetics only.** All conclusions come only from Sérgio F genes (without blood tests or exams). They are predispositions, non-diagnosis. Each theme is divided into favourable and unfavourable; what does not have clear judgment appears as typical.

**Strand:** All genotypes are in the strand + of dbSNP (GRCh38). In the SNPs of this report, 15 directly agree with the dbSNP alleles, 0 needed conversion to the complementary strand and 0 are palindromic (A/T or C/G; orientation confirmed by the GRCh38 position). The genotypes are already in the strand + of the reference. Therefore, some genotypes appear with different letters from those used in the literature when using the strand of the gene (e.g. MTHFR C677T = rs1801133 A in the strand +).

**Origin and trust:** chip = directly genotyped (high confidence); imputation DR2=x = locally inferred from a public panel of complete genomes (30x), accepted only with  $DR2 \geq 0.8$  (good trust, just below the chip); SNP not found in the data : neither chip nor imputed with sufficient quality (indels, rare variants, deletions).

**Effect and judgment allele:** The comparison between the number of Sérgio F copies and the mean of the reference population ( $2 \times$  allele frequency; column «Freq.»). **Unfavourable** = at least 0.6 copies of risk allele above average (or any copy of a rare variant of strong effect); **Favourable** = at least 0.6 below average, protective allele, or absence of a strong effect variant; **Typical / no judgement** = close to the average (e.g. homozygous for a risk allele that is the most common) or context variant. Each common SNP has a small individual effect; what weighs more are the variants of large effect (marked as strong evidence) and the polygenic scores.

**Polygenic scores (PGS):** add thousands of variants with coefficients published in PGS Catalog. The percentile compares Sérgio F with the reference population (n=633), calculated with exactly the same variants. Percentage  $\geq 80$  in a risk score = unfavourable;  $\leq 20$  = favourable.

## Polygenic scores

The testosterone and SHBG scores (Ding 2023, 980 thousand variants, specific to man) are different from those used in the report "Hormones" (which gave a percentile 58 and 66 with smaller scores): both agree in the direction (above the average). Here, high genetic testosterone counts as favourable for andropause; SHBG, LH, FSH and puberty age are left without judgment.

| Trait                                      | Score     | Variables used (coverage) | Percentage (reference population) | Judgement              |
|--------------------------------------------|-----------|---------------------------|-----------------------------------|------------------------|
| Testosterone in man (979 739 variants)     | PGS003559 | 972 627 of 979 739 (99%)  | p89                               | Favourable             |
| SHBG (979 739 variants)                    | PGS003552 | 972 627 of 979 739 (99%)  | p87                               | Typical / no judgement |
| LH (GWAS Catalog score, 6 loci)            | C_LH      | 4 of 6 (67%)              | p76                               | Typical / no judgement |
| FSH (GWAS Catalog score, 5 loci)           | C_FSH     | 5 of 5 (100%)             | p12                               | Typical / no judgement |
| Loss of mosaic Y chromosome (114 variants) | PGS003575 | 111 of 114 (97%)          | p37                               | Typical / no judgement |
| Age of puberty (voice; 31 loci)            | C_VOZ     | 29 of 31 (94%)            | p88                               | Typical / no judgement |

# Hypothalamic-pituitary-testicle and suprarenal axis (SHBG, LH, DHEA)

LH stimulates testosterone production; SHBG carries it in the blood and defines the free fraction. DHEA and DHEA-S (suprarenal) fall ~2% per year.

**Polygenic scores of this theme:** Testosterone in man (979 739 variants) (PGS003559): the 89th percentile **Favourable** ; SHBG (979 739 variants) (PGS003552): the 87th percentile **Typical / no judgement** ; LH (GWAS Catalog score, 6 loci) (C\_LH): the 76th percentile **Typical / no judgement** ; FSH (GWAS Catalog score, 5 loci) (C\_FSH): the 12th percentile **Typical / no judgement**

| Gene          | SNP        | Genotype (+) | Source | Effect allele | Copies | Freq. | Effect                                                                                                                                | Judgement              |
|---------------|------------|--------------|--------|---------------|--------|-------|---------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| SHBG          | rs12150660 | GT           | chip   | T             | 1      | 0.24  | SHBG (promoter/intron): T allele low to SHBG; SHBG rises with age and is the main driver of free testosterone drop. (strong evidence) | Typical / no judgement |
| SHBG          | rs727428   | TC           | chip   | C             | 1      | 0.57  | SHBG 3' SHBG and total testosterone. (moderate evidence)                                                                              | Typical / no judgement |
| GCKR          | rs780094   | TC           | chip   | C             | 1      | 0.59  | GCKR (liver): Lower SHBG with the allele that rises the triglycerides. (moderate evidence)                                            | Typical / no judgement |
| LHCGR N312S   | rs2293275  | CC           | chip   | —             | —      | —     | LH receptor in Leydig cells: common polymorphism. (weak evidence)                                                                     | Typical / no judgement |
| FSHB          | rs10835638 | GT           | chip   | T             | 1      | 0.14  | FSHB -211G>T: less FSH and more LH; T is associated with a slightly higher testosterone and smaller testes. (moderate evidence)       | Typical / no judgement |
| CYP17A1/NT5C2 | rs11191548 | TT           | chip   | —             | —      | —     | CYP17A1 (17-hydroxylase; DHEA): adrenal and testicular steroidogenesis. (weak evidence)                                               | Typical / no judgement |
| SULT2A1       | rs182420   | TT           | chip   | —             | —      | —     | DHEA sulphotransferase: DHEA-S circulating (falls ~2% per year after 30). (moderate evidence)                                         | Typical / no judgement |
| ZKSCAN5       | rs11761528 | CC           | chip   | —             | —      | —     | ZKSCAN5/CYP3A7: DHEA-S. (moderate evidence)                                                                                           | Typical / no judgement |

**Favourable**

- P89 male testosterone
- SHBG rs12150660 GT (1 allele of SHBG low)

**Unfavourable**

- SHBG polygenic p87

**Interpretation:** High + high SHBG = free testosterone probably close to the average. FSHB -211 GT: less FSH, a little more LH; FSH score at the 12th percentile and LH in the 76 (this one with low coverage). GCKR and SHBG rs727428 heterozygotes. The genes of DHEA (SULT2A1, ZKSCAN5, CYP17A1) with no effect on the data; the genetic DHEA-S is in the report "Hormones".

# Y-chromosome loss with age

With age, a growing fraction of blood cells loses the Y chromosome (mLOY). At 70 it affects ~40% of men to some degree and is associated with cancer, Alzheimer's disease, cardiac fibrosis and mortality. Tobacco is the main environmental factor.

**Polygenic scores of this theme:** Loss of mosaic Y chromosome (114 variants) (PGS003575): the 37th percentile Typical / no judgement

| Gene           | SNP         | Genotype (+) | Source                    | Effect allele | Copies | Freq. | Effect                                                                                                      | Judgement                 |
|----------------|-------------|--------------|---------------------------|---------------|--------|-------|-------------------------------------------------------------------------------------------------------------|---------------------------|
| TCL1A          | rs2887399   | GG           | chip                      | T             | 0      | 0.21  | TCL1A: T allele reduces the loss of Y chromosome in blood cells and clonal hematopoiesis. (strong evidence) | Typical / no judgement    |
| SETBP1         | rs17758695  | CC           | chip                      | —             | —      | —     | SETBP1: loss of the Y chromosome in mosaic. (moderate evidence)                                             | Typical / no judgement    |
| TERT           | rs7705526   | CA           | chip                      | A             | 1      | 0.35  | TERT: Allele A (long telomeres) increases the loss of Y and CHIP. (moderate evidence)                       | Typical / no judgement    |
| CHEK2 1100delC | rs555607708 |              | SNP not found in the data | —             | —      | —     | CHEK2 1100delC: greatly increases the loss of Y and the risk of several cancers (rare). (strong evidence)   | SNP not found in the data |

**Favourable**

- MLOY score at the 37th percentile

**Unfavourable**

- TCL1A rs2887399 GG (without protection)
- TERT rs7705526 CA

**Interpretation:** Typical predisposition, low side. Without tobacco (the biggest environmental accelerator), the real risk is below the genetic. SETBP1 CC without defined effect; CHEK2 1100delC not visible.

# Reproductive calendar: age of puberty

The age of puberty (voice change) marks the reproductive calendar. Early puberty is associated with more cardiometabolic risk; later bone density is slightly lower.

**Polygenic scores of this theme:** Age of puberty (voice; 31 loci (C\_VOZ): the 88th percentile

Typical / no judgement

| Gene         | SNP        | Genotype (+) | Source | Effect allele | Copies | Freq. | Effect                                                                | Judgement              |
|--------------|------------|--------------|--------|---------------|--------|-------|-----------------------------------------------------------------------|------------------------|
| LIN28B       | rs314276   | AC           | chip   | —             | —      | —     | LIN28B: the strongest locus of puberty age (voice). (strong evidence) | Typical / no judgement |
| MKRN3/MAGEL2 | rs12148769 | GG           | chip   | —             | —      | —     | 15q11 (printing): age of puberty. (weak evidence)                     | Typical / no judgement |
| TMEM18       | rs2947411  | GG           | chip   | —             | —      | —     | TMEM18: puberty and adiposity. (weak evidence)                        | Typical / no judgement |

**Favourable**

- None.

**Unfavourable**

- None.

**Interpretation:** Score of the age of voice change at the 88th percentile: genetically late puberty (profile associated with less cardiometabolic risk). LIN28B AC, MKRN3 GG and TMEM18 GG. No judgment.

# Action in tissues: estrogen receptors and androgens

Part of the action of testosterone in bone, brain and fat involves conversion into estradiol (aromatase) and estrogen receptor; the sensitivity of the androgen receptor depends on the repetition CAG.

| Gene   | SNP       | Genotype (+) | Source                 | Effect allele | Copies | Freq. | Effect                                                                                                                                   | Judgement              |
|--------|-----------|--------------|------------------------|---------------|--------|-------|------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| ESR1   | rs2234693 | TT           | chip                   | —             | —      | —     | ESR1 PvuII: most of the action of testosterone in the bone and in the male brain passes through estradiol (via ESR1). (weak evidence)    | Typical / no judgement |
| AR (X) | rs5919392 | C            | imputation<br>DR2=0.98 | —             | —      | —     | Androgen receptor: the CAG repeat (receptor sensitivity) is not read by the chip; this SNP is just a marker of the gene. (weak evidence) | Typical / no judgement |

Favourable

• None.

Unfavourable

• None.

**Interpretation:** ESR1 PvuII TT and AR (marker) C: no effect defined. The CAG repetition is not read by the chip. See 'Hormones' for CYP19A1 and estradiol.

## Mechanisms of ageing

Each finding was linked to the aging mechanisms of the list of objectives for immortality (84 mechanisms). The table shows only the mechanisms played by this report.

| Mechanism                   | Unfavourable                | Favourable                                                | Interpretation                              |
|-----------------------------|-----------------------------|-----------------------------------------------------------|---------------------------------------------|
| Endocrine dysregulation     | High polygenic SHBG         | P89 male testosterone                                     | Androgen reserve above average.             |
| Sarcopenia / Thin Mass Loss | —                           | High genetic testosterone                                 | Testosterone maintains muscle and strength. |
| Osteoporosis                | Late puberty (p88)          | High testosterone (converted into estradiol in the bone)  | Neutral.                                    |
| Insulin resistance          | —                           | High SHBG (protects glucose) and high testosterone in man | Favourable.                                 |
| Somatic / Tissue Mosaicism  | TCL1A GG                    | MLOY p37 score                                            | Typical.                                    |
| Clonal hematopoiesis (CHIP) | TCL1A GG; TERT rs7705526 CA | —                                                         | Typical.                                    |
| Immunosenescence            | —                           | Low-typical genetic mLOY                                  | Neutral.                                    |

## Practical suggestions

All the suggestions below are suggestion (not applied): only from genetics, they are aligned with the v2 genetic plan and do not change the current stack without Sérgio F decision and the doctor.

| Topic                               | Suggestion (not applied)                                                                                                                                                                             | Genetic basis                                             |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| Sleep                               | 7-8 h per night: one week at 5 h low testosterone 10-15%. Melatonin as in the plan (0.3-1 mg). <b>suggestion (not applied)</b>                                                                       | High genetic reserve; high SHBG                           |
| Strength training                   | 2-4 sessions per week with multiarticular exercises; protects testosterone and lean mass. <b>suggestion (not applied)</b>                                                                            | Andropause                                                |
| Visceral Adiposity                  | The reduction of visceral adiposity (training, retalutride) is the most effective way to increase total and free testosterone. <b>suggestion (not applied)</b>                                       | High SHBG                                                 |
| Rapamycin                           | MTOR inhibitors can reduce testosterone; if symptoms (energy, libido) arise, review the regimen with the physician. <b>suggestion (not applied)</b>                                                  | High reserve gives margin, but does not cancel the effect |
| No exogenous testosterone by defect | Do not start testosterone without confirmed hypogonadism and symptoms: suppresses its own production and fertility. <b>suggestion (not applied)</b>                                                  | High genetic reserve                                      |
| DHEA and vitamin D                  | DHEA: it is not necessary to supplement (see 'Hormones'). D3 10 000 IU and omega-3 3.2-4 g/day; zinc by food. <b>suggestion (not applied)</b>                                                        | Coordination with the plan                                |
| Chromosome Y                        | Zero tobacco is the main measure against Y loss; exercise and less inflammation also help. <b>suggestion (not applied)</b>                                                                           | TCL1A GG                                                  |
| What not to do                      | Without telomerase activators (the TERT allele of long telomeres already increases the CHIP). Curcumin, luteolin, apigenin and quercetin in pause while ALT is high. <b>suggestion (not applied)</b> | V2 plan rules                                             |

## Limitations

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- The CAG repeat of the androgen receptor, the copy variation and the rare variants of the axis (KISS1R, GNRHR, LHB) are not read by the chip.
- The LH and FSH scores use 5–6 variants of the GWAS Catalog (the LH with a weighted coverage of only 20%).
- The percentiles compare with the reference population (n=633).
- Only genes: does not use hormonal analysis.

## Methods and sources

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- Genotypes: Raw DNA of the chip (GRCh38, strand +), analyzed only locally; Missing SNPs (statistical imputation, public reference panel of complete 30x genomes), accepted with  $DR2 \geq 0.8$ .
- Public positions and alleles: Ensembl REST (only rsIDs). Locally calculated frequencies in the reference population (n=633).
- Scores: PGS003559 (testosterone in man) and PGS003552 (SHBG), Ding 2023; PGS003575 (loss of the mosaic Y chromosome); simple scores from GWAS Catalog (local copy) to LH (6 loci), FSH (5) and age of male puberty (31). Percentages vs. reference population (n=633) with exactly the same variants.
- Main references: Harman 2001 and Wu 2008 (decline of testosterone with age); Ohlsson 2011 and Ruth 2020 (genetics of testosterone and SHBG, PLoS Genet / Nat Med); Thompson 2019 (loss of Y, Nature); Weinstock 2023 (TCL1A); Day 2015 and Hollis 2020 (male puberty); Leproult 2011 (sleep and testosterone, JAMA); Huyghe 2007 (mTOR inhibitors and male gonads).
- Mapping for 84 aging mechanisms of the target list. Report generated locally (HTML → PDF).